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Development and validation of a nomogram prediction model for coronary heart disease in diabetic patients: a study based on the 2011–2020 NHANES database

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Abstract

Objective To analyze the risk and influencing factors for coronary heart disease (CHD) in patients with diabetes (DM), and to develop and validate a nomogram prediction model, providing a basis for the early diagnosis and individualized intervention in patients with DM and CHD.

Methods This study was based on data from the National Health and Nutrition Examination Survey (NHANES). A total of 2,141 diabetic patients from 2011 to 2020 were included, randomly divided into a training set ($n=1,499$) and a validation set ($n=642$) at a 7:3 ratio. The least absolute shrinkage and selection operator (Lasso) regression analysis was used to screen risk factors, and a multivariate logistic regression model was developed to construct the DM-CHD nomogram prediction model. Model performance was internally validated using receiver operating characteristic (ROC) curves, calibration curves, and decision curve analysis (DCA).The Hosmer–Lemeshow Test was utilized to evaluate the overall goodness of fit of the nomogram.

Results Univariate analysis identified 15 factors as risk factors for DM-related CHD. Lasso regression further selected 7 key predictors: Age (OR 1.06, CI 1.05–1.08, $P<0.001$), Gender (OR 0.47, CI 0.36–0.63, $P<0.001$), Hypertension (OR 1.85, CI 1.33–2.57, $P<0.001$), Weight Adjusted Waist Index (OR 1.50, CI 1.25–1.81, $P<0.001$), Neutrophils (OR 1.09, CI 1.02–1.17, $P=0.009$), Platelets (OR 0.99, CI 0.99–0.99, $P<0.001$), and Triglycerides (OR 1.18, CI 1.08–1.30, $P<0.001$). The area under the ROC curve (AUC) for the nomogram model was 0.758 (95% CI 0.728–0.789) in the training set and 0.747 (95% CI 0.699–0.796) in the validation set. Calibration curves and DCA indicated that the model exhibited satisfactory predictive performance. The model's reliability and clinical net benefit were further validated.

Conclusion The nomogram model developed in this study, based on multiple clinical indicators (Age, Gender, Hypertension, Weight Adjusted Waist Index, Neutrophils, Platelets, and Triglycerides), demonstrated adequate calibration and clinical net benefit in the validation cohort. The model demonstrates moderate but clinically useful discrimination ability, providing scientific guidance for early diagnosis and personalized interventions in patients with DM complicated by CHD, and may help reduce CHD risk in diabetic patients.

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Keywords Coronary Heart Disease, Diabetes, NHANES

Background

Based on NHANES data from 2017 to March 2020, the prevalence of cardiovascular disease (CVD) in adults aged ≥ 20 years was 48.6% [1]. The World Health Organization (WHO) predicts that by 2030, nearly 9.3 million people will die annually from CVD [2]. Coronary heart disease (CHD), as a major component of CVD, affects 244.11 million people worldwide, resulting in a significant disease burden [3]. Meanwhile, by 2021, approximately 529 million people worldwide were affected by diabetes (DM) [4], and it is estimated that by 2030, 552 million people will suffer from DM [5]. CVD is the primary cause of morbidity and mortality among diabetic patients [6]. Due to the presence of neuropathy and other metabolic abnormalities, DM patients often lack early symptoms and are usually diagnosed only when atherosclerosis and its complications (such as CHD) occur, at which point the disease is often irreversible [7].

Although existing studies offer some value in predicting cardiovascular risk in DM patients, they have several limitations: First, many studies have small sample sizes or are limited to specific populations (such as a single ethnic groups or regions), restricting the generalizability of their findings [8, 9]. Second, insufficient attention has been paid to certain potential biomarkers (such as Weight Adjusted Waist Index, Neutrophil, Platelet, etc.), which may play a crucial role in the development of CHD. Additionally, current models often rely on expensive imaging or specialized laboratory tests, limiting their applicability in routine clinical practice.

In the context of precision medicine, this study aims to develop and validate a nomogram-based CHD risk prediction model for DM patients, using multidimensional clinical indicators from the widely representative NHANES database. By incorporating clinical data and biomarkers that are easily accessible in routine clinical practice (such as Age, Gender, Hypertension, Weight Adjusted Waist Index, Neutrophils, Platelets, Triglycerides, High-Density Lipoprotein Cholesterol, Low-Density Lipoprotein, etc.), this study not only fills existing gaps in key predictive factors but also provides a nomogram to visualize the contribution of each risk factor. This will offer clinicians a simple and efficient tool for early risk stratification and individualized intervention in patients with DM, ultimately reducing cardiovascular risk.

Data and methods

Data collection

This study utilized data from the 2011–2020 National Health and Nutrition Examination Survey (NHANES) database, which collects information on the health, nutritional status, and health behaviors of non-institutionalized civilian residents of the United States through personal interviews, standardized physical exams, and laboratory tests [10]. The NHANES database provides a diverse sample, ensuring the broad applicability of the study results.

Inclusion and exclusion criteria

Inclusion criteria: 1

Participants who answered "YES" to the question, "Doctor told you have diabetes?" (DIQ010), in the 2011–2020 NHANES database multiple-choice questionnaire were classified as DM participants. 2. Participants who answered "YES" to the question, "Has a doctor or other health professional told you that you have CHD?" (MCQ160C), in the 2011–2020 NHANES database multiple-choice questionnaire were classified into the CHD group, while those who answered "NO" were classified into the non-CHD group.

Exclusion criteria: 1

Participants with missing information regarding DM or CHD were excluded. 2. Participants with a missing data proportion of 15% or greater for any of the variables intended for analysis were excluded.

A total of 2,141 DM patients were included in the study, of whom 379 patients were diagnosed with CHD. All eligible participants were randomly assigned to a training set (n = 1,499) and a validation set (n = 642) in a 7:3 ratio. The training and validation sets were further divided into CHD and non-CHD groups based on the presence of CHD.

Variables included in the analysis

The following clinical data were selected as potential risk factors for analysis:

1. Demographic data: Age, Gender, Race, Education level.
2. Past Medical history: Hypertension (HT), Take medication for depression, Feelings of tiredness or lack of energy, and Sleep difficulties.
3. Laboratory data: Body Mass Index (BMI), Weight Adjusted Waist Index (WWI) $= WaistCircumference(cm)/\sqrt{Weight(kg)}$, Hypertension (HT), Systolic Blood Pressure (SBP), Diastolic Blood Pressure (DBP), Pulse Pressure (PP), Aspartate Aminotransferase(AST), Alanine Aminotransferase (ALT), Creatinine (CR), Neutrophils (NEU), Lymphocytes (LYM), Platelets (PLT), High-Density Lipoprotein Cholesterol (HDL-C), Low-density Lipoprotein (LDL), Total Cholesterol (TC), Fasting Glucose(GLU), Triglyceride (TG),Triglyceride-GlucoseIndex(TyG)

$= ln[TG(mg/dl) * FBG(mg/dl)/2]$. All glucose values mentioned are fasting measurements.

Statistical analysis

The research data were analyzed using R statistical software (version 4.2.2). Normally distributed continuous variables were described using the mean ± standard deviation (mean ± SD), while non-normally distributed continuous variables were described using the median and interquartile range (median (IQR)). Intergroup comparisons were conducted using the Mann–Whitney U test. Categorical variables were described using counts and percentages (n (%)), and intergroup comparisons were performed using the chi-square test (χ^2).

Predictor Variable Selection for Model Inclusion: Univariate, multivariate logistic regression, and Lasso regression analyses were used to identify variables significantly associated with CHD, avoiding multicollinearity and identifying key predictors. Lasso regression utilized cross-validation to determine the optimal λ value and select statistically significant variables, thereby identifying independently associated risk factors.

Model Construction and Validation: Based on the results of multivariate logistic regression analysis, a nomogram prediction model was developed to predict the probability of CHD in DM patients. The model was internally validated using the following methods: the area under the ROC curve (AUC) for the training and validation sets to evaluate model discrimination ability; the Calibration curve to assess the consistency between predicted probabilities and actual outcomes; and the DCA curve to evaluate clinical net benefit at different threshold probabilities. The Hosmer–Lemeshow Test was utilized to evaluate the overall goodness of fit of the nomogram.The performance of the nomogram was further evaluated in the validation and test dataset.

Results

Baseline characteristics of the two groups

A total of 2,141 DM participants were included in this study, of whom 379 had CHD. Participants were randomly assigned to a training set ($n = 1{,}499$) and a validation set ($n = 642$) in a 7:3 ratio (Fig. 1).

Table 1 lists the baseline characteristics of patients in the training and validation sets. The results showed that the demographic characteristics, medical history, and

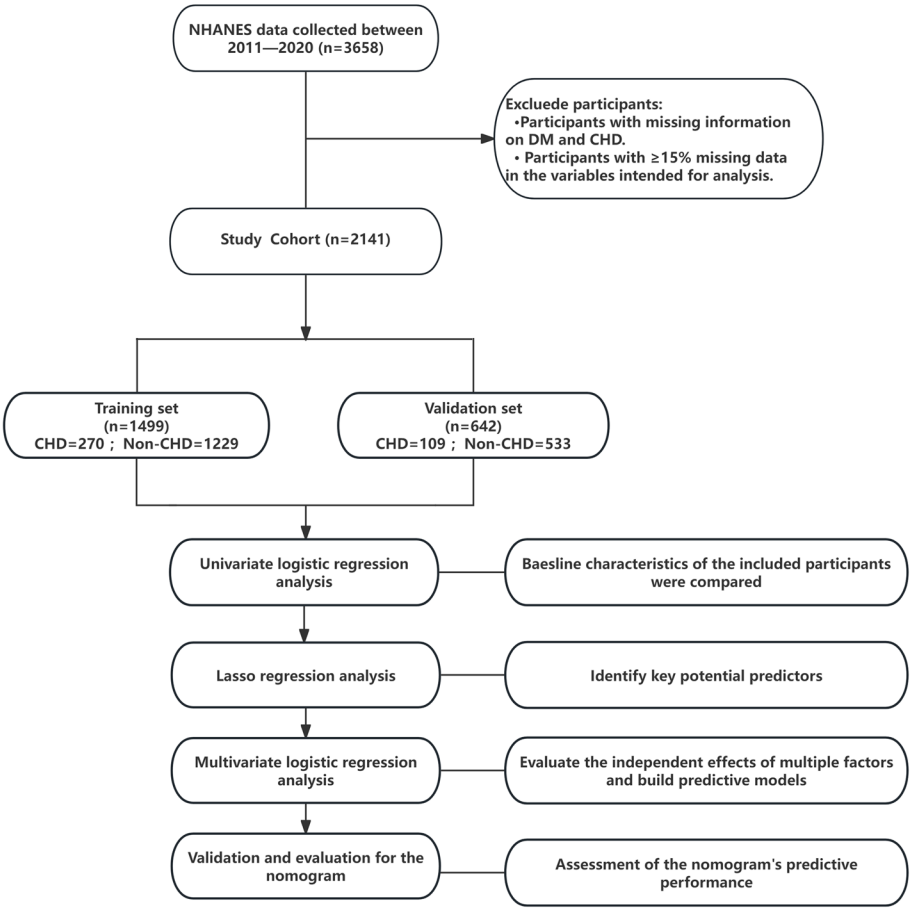


Fig. 1 Flowchart. National Health and Nutrition Examination Survey (NHANES), Diabetes (DM), Coronary Heart Disease (CHD)

Table 1 Basic characteristics of patients in the derivation set and validation set

		Cohort		
Characteristics	**Overall, N = 2,141**	**Training Cohort, N = 1,499**	**Internal Test Cohort, N = 642**	***p*-value**
Demographics				
Age Median (IQR)	63 (54, 72)	64 (54, 72)	63 (54, 72)	0.481
Gender				0.84
Male	1,151 (53.8%)	808 (53.9%)	343 (53.4%)	
Female	990 (46.2%)	691 (46.1%)	299 (46.6%)	
Race				0.398
Mexican American	302 (14.1%)	219 (14.6%)	83 (12.9%)	
Non-Hispanic Asian	413 (19.3%)	301 (20.1%)	112 (17.4%)	
Non-Hispanic White	291 (13.6%)	198 (13.2%)	93 (14.5%)	
Non-Hispanic Black	841 (39.3%)	571 (38.1%)	270 (42.1%)	
Other Hispanic	222 (10.4%)	159 (10.6%)	63 (9.8%)	
Other Race—Including Multi-Racial	72 (3.4%)	51 (3.4%)	21 (3.3%)	
Education level				0.872
Less than 9th grade	275 (12.8%)	195 (13.0%)	80 (12.5%)	
9-11th grade (Includes 12th grade with no diploma)	279 (13.0%)	195 (13.0%)	84 (13.1%)	
High school graduate/GED or equivalent	492 (23.0%)	337 (22.5%)	155 (24.1%)	
Some college or AA degree	668 (31.2%)	476 (31.8%)	192 (29.9%)	
College graduate or above	427 (19.9%)	296 (19.7%)	131 (20.4%)	
BMI Median (IQR)	31 (27, 36)	31 (27, 36)	31 (28, 37)	0.257
WWI Mean ± SD	11.67 ± 0.74	11.66 ± 0.72	11.70 ± 0.77	0.233
Past medical history				
HT				0.478
NO	611 (28.5%)	421 (28.1%)	190 (29.6%)	
YES	1,530 (71.5%)	1,078 (71.9%)	452 (70.4%)	
SBP Median (IQR)	128 (120, 136)	128 (120, 136)	128 (120, 135)	0.621
DBP Median (IQR)	77 (70, 80)	77 (70, 80)	77 (70, 80)	0.911
PP Median (IQR)	52 (45, 60)	52 (45, 60)	52 (44,60)	0.659
Take medication for depression				0.624
NO	1,892 (88.4%)	1,328 (88.6%)	564 (87.9%)	
YES	249 (11.6%)	171 (11.4%)	78 (12.1%)	
Feeling tired or having little energy				0.921
NO	1,687 (78.8%)	1,182 (78.9%)	505 (78.7%)	
YES	454 (21.2%)	317 (21.1%)	137 (21.3%)	
Trouble sleeping or sleeping too much				0.903
NO	1,704 (79.6%)	1,192 (79.5%)	512 (79.8%)	
YES	437 (20.4%)	307 (20.5%)	130 (20.2%)	
Laboratory parameters				
ALT(U/L) Median (IQR)	20 (15, 28)	20 (15, 28)	21 (15, 28)	0.401
AST(U/L) Median (IQR)	21 (17, 26)	21 (17, 26)	21 (17, 27)	0.214
CR(umol/L) Median (IQR)	80 (65, 102)	80 (65, 103)	79 (64, 99)	0.309
NEU(1000 cell/uL) Median (IQR)o	4.30 (3.40, 5.50)	4.30 (3.40, 5.50)	4.30 (3.50, 5.50)	0.907
LYM(1000 cells/uL) Median (IQR)	2.00 (1.60, 2.60)	2.00 (1.60, 2.60)	2.00 (1.70, 2.70)	0.159
PLT(1000 cells/uL) Median (IQR)	226 (189, 272)	226 (189, 269)	228 (188, 277)	0.251
HDL-C(mmol/L) Median (IQR)	7.5 (5.9, 8.9)	7.5 (5.9, 9.0)	7.4 (5.5, 8.8)	0.148
LDL(mg/dL) Median (IQR)	2 (1, 10)	2 (1, 9)	2 (1, 10)	0.476
TC(mmol/L) Median (IQR)	3.0 (1.2, 5.4)	2.8 (1.2, 5.3)	3.4 (1.4, 5.7)	0.008
GLU(mmol/L) Median (IQR)	7.00 (6.50, 7.86)	7.00 (6.50, 7.83)	7.04 (6.50, 7.88)	0.378
TG(mmol/L) Median (IQR)	1.29 (1.03, 1.76)	1.29 (1.03, 1.78)	1.24 (1.01, 1.68)	0.08
TyG Median (IQR)	8.87 (8.68, 9.18)	8.87 (8.68, 9.19)	8.85 (8.67, 9.15)	0.177

Data are expressed as means ± standard deviations, medians with interquartile ranges, or frequencies and percentages

Abbreviations: *BMI* Body Mass Index, *WWI* Weight Adjusted Waist Index, *HT* Hypertension, *SBP* Systolic Blood Pressure, *DBP* Diastolic Blood Pressure, *PP* Pulse Pressure, *ALT* Alanine Aminotransferase, *AST* Aspartate Aminotransferase, *CR Creatinine*, *NEU* Neutrophils, *LYM* Lymphocytes, *PLT* Platelets, *HDL-C* High-Density Lipoprotein Cholesterol, *LDL* Low-Density Lipoprotein, *TC* Total Cholesterol, *GLU* Fasting Glucose, *TG* Triglycerides, *TyG* Triglyceride-Glucose Index

laboratory test indicators of the training and validation sets were comparable, with no statistically significant differences for any variable except for TC, indicating that the random allocation of the dataset was reasonable and appropriate for subsequent analysis.

Univariate analysis

Univariate analysis was conducted to compare the baseline characteristics of the CHD and non-CHD groups, ensuring balanced subgroup sample sizes and minimizing potential bias from unequal sample distributions. Table 2 presents the results of the univariate analysis. In the training set, 15 factors were identified as risk factors for DM-related CHD ($P<0.05$). Demographic data and Past medical history indicated that, compared to the non-CHD group, the CHD group had older age, a greater proportion of males, a higher prevalence of hypertension, and a higher proportion of individuals reporting fatigue, lack of energy, and sleep difficulties ($P<0.05$). Laboratory results showed that, compared to the non-CHD group, the CHD group had higher WWI, PP, CR, NEU, TG, and TyG levels, while LYM, PLT, and DBP were lower ($P<0.05$).

Model construction

Selection of predictors

Lasso regression analysis was performed on the above-mentioned factors with significant differences to identify 7 key potential predictors: Age, Gender, HT, WWI, NEU, PLT, and TG ($\lambda = 0.0225701180533294$). These 7 key predictors were incorporated into the model development in the next step (Fig. 2).

Univariate logistic regression analysis

The 7 key potential predictors identified by Lasso regression analysis were further analyzed using univariate logistic regression (Table 3). Age (OR 1.06, CI 1.05–1.08, $P<0.001$), Gender (OR 0.47, CI 0.36–0.63, $P<0.001$), HT (OR 1.85, CI 1.33–2.57, $P<0.001$), WWI (OR 1.50, CI 1.25–1.81, $P<0.001$), NEU (OR 1.09, CI 1.02–1.17, $P=0.009$), PLT (OR 0.99, CI 0.99–0.99, $P<0.001$), and TG (OR 1.18, CI 1.08–1.30, $P<0.001$) were all significantly associated with the risk of CHD in DM patients, with each association being statistically significant.

Multivariate logistic regression analysis

The potential predictors identified by Lasso regression were included in the multivariate logistic regression model (Table 4). The results showed that Age (OR 1.05, CI 1.03–1.07, $P<0.001$), Gender (OR 0.43, CI 0.31–0.60, $P<0.001$), HT (OR 1.50, CI 1.06–2.13, $P=0.024$), WWI (OR 1.47, CI 1.17–1.84, $P<0.001$), NEU (OR 1.15, CI 1.06–1.25, $P<0.001$), PLT (OR 0.99, CI 0.99–1.00, $P<0.001$), and TG (OR 1.27, CI 1.14–1.41, $P<0.001$) were identified as independent predictors of CHD in DM patients.

Nomogram model construction

Based on the results of the multivariate logistic regression analysis, a nomogram prediction model was developed (Fig. 3). This model converts the contribution of each predictor into a score, and by summing the scores of all variables, calculates the total risk score for CHD in DM patients.

Model evaluation and validation

The AUC of the nomogram model was 0.758 (95% CI 0.728–0.789) in the training set and 0.747 (95% CI 0.699–0.796) in the validation set (Fig. 4B), indicating that the nomogram model developed in this study has satisfactory predictive performance. Additionally, the AUC values for individual factors were all greater than 0.5, with Age (AUC = 0.681, 95% CI 0.649–0.714), Gender (AUC = 0.589, 95% CI 0.558–0.620), HT (AUC = 0.556, 95% CI 0.529–0.583), NEU (AUC = 0.543, 95% CI 0.506–0.581), PLT (AUC = 0.643, 95% CI 0.606–0.679), TG (AUC = 0.554, 95% CI 0.514–0.594), and WWI (AUC = 0.586, 95% CI 0.549–0.623) (Fig. 4A). Although the predictive ability of individual factors was limited, their combined use significantly enhanced the overall predictive performance of the model.

Calibration curve

The calibration curve (Fig. 5) shows a high degree of agreement between the predicted and observed probabilities in both the training and validation sets. The calibration curve for the training set is close to the ideal line. The calibration curve for the validation set also demonstrates good fit, indicating strong consistency between the predicted probabilities and actual outcomes, further validating the model's reliability and calibration performance.

DCA curve analysis

The decision curve analysis (DCA, Fig. 6) shows that the prediction model provides a high clinical net benefit over a wide range of threshold probabilities (11%–93% for the training set and 8%–100% for the validation set). Compared to the "All-None" strategy (assuming all patients are low risk), the model curve is above the "All-None" curve across most of the cost-effectiveness range, suggesting that the model offers a higher clinical net benefit.

In the training set, the model curve reaches its peak net benefit at a high-risk threshold of approximately 0.2. In the validation set, the peak net benefit threshold is slightly above 0.2, indicating that, in practical applications, the high-risk threshold could be set around 0.2.

Simultaneously, the Hosmer–Lemeshow goodness-of-fit test showed that the p-value was greater than 0.05 for

Table 2 Findings among patients with DM who did or did not develop CHD in training set and validation set

Characteristics		Training Cohort		Internal Test Cohort		
	NO, *N* = 1,229	YES, *N* = 270	*p*-value	NO, *N* = 533	YES, *N* = 109	*p*-value
Demographics						
Age Median (IQR)	62 (53, 71)	70 (63, 76)	< 0.001	62 (53, 70)	70 (62, 76)	< 0.001
Gender			< 0.001			0.065
Male	623 (51%)	185 (69%)		276 (52%)	67 (61%)	
Female	606 (49%)	85 (31%)		257 (48%)	42 (39%)	
Race			0.015			0.292
Mexican American	193 (16%)	26 (10%)		73 (14%)	10 (9%)	
Non-Hispanic Asian	257 (21%)	44 (16%)		97 (18%)	15 (14%)	
Non-Hispanic White	152 (12%)	46 (17%)		72 (14%)	21 (19%)	
Non-Hispanic Black	454 (37%)	117 (43%)		218 (41%)	52 (48%)	
Other Hispanic	132 (11%)	27 (10%)		55 (10%)	8 (7%)	
Other Race—Including Multi-Racial	41 (3%)	10 (4%)		18 (3%)	3 (3%)	
Education level			0.83			0.989
Less than 9th grade	161 (13%)	34 (13%)		65 (12%)	15 (14%)	
9-11th grade (Includes 12th grade with no diploma)	154 (13%)	41 (15%)		70 (13%)	14 (13%)	
High school graduate/GED or equivalent	276 (22%)	61 (23%)		129 (24%)	26 (24%)	
Some college or AA degree	394 (32%)	82 (30%)		161 (30%)	31 (28%)	
College graduate or above	244 (20%)	52 (19%)		108 (20%)	23 (21%)	
BMI Median (IQR)	31 (27, 36)	31 (28, 36)	0.758	31 (28, 37)	31 (28, 36)	0.312
WWI Mean ± SD	1.62 ± 0.72	11.83 ± 0.72	< 0.001	11.66 ± 0.77	11.90 ± 0.75	0.003
Past medical history						
HT			< 0.001			0.001
NO	370 (30%)	51 (19%)		172 (32%)	18 (17%)	
YES	859 (70%)	219 (81%)		361 (68%)	91 (83%)	
SBP Median (IQR)	128 (121, 136)	129 (120, 136)	0.605	128 (120, 136)	128 (120, 135)	0.165
DBP Median (IQR)	77 (72, 80)	75 (69, 80)	< 0.001	78 (72, 80)	74 (68, 80)	0.041
PP Median (IQR)	51(45,60)	55(48,61)	0.002	51(46,60)	52(46,60)	0.746
Take medication for depression			0.272			0.572
NO	1,094 (89%)	234 (87%)		470 (88%)	94 (86%)	
YES	135 (11%)	36 (13%)		63 (12%)	15 (14%)	
Feeling tired or having little energy			0.022			0.006
NO	983 (80%)	199 (74%)		430 (81%)	75 (69%)	
YES	246 (20%)	71 (26%)		103 (19%)	34 (31%)	
Trouble sleeping or sleeping too much			0.009			0.07
NO	993 (81%)	199 (74%)		432 (81%)	80 (73%)	
YES	236 (19%)	71 (26%)		101 (19%)	29 (27%)	
Laboratory parameters						
AST(U/L) Median (IQR)	21 (17, 26)	21 (18, 26)	0.278	21 (17, 27)	21 (17, 27)	0.967
CR(umol/L) Median (IQR)	77 (64, 98)	95 (78, 117)	< 0.001	76 (63, 94)	95 (78, 118)	0.053
NEU (1000 cell/uL) Median (IQR)	4.30 (3.40, 5.50)	4.40 (3.53, 5.80)	0.014	4.30 (3.40, 5.40)	4.90 (3.70, 5.90)	0.004
LYM (1000 cells/uL) Median (IQR)	2.10 (1.60, 2.60)	1.90 (1.50, 2.40)	0.005	2.10 (1.70, 2.70)	1.80 (1.50, 2.20)	0.001
PLT(1000 cells/uL) Median (IQR)	230 (194, 279)	202 (164, 241)	< 0.001	234 (193, 280)	200 (170, 246)	< 0.001
HDL-C(mmol/L) Median (IQR)	7.6 (5.9, 8.9)	7.4 (6.0, 9.0)	0.705	7.4 (5.6, 8.8)	7.4 (1.7, 9.1)	0.527
LDL(mg/dL) Median (IQR)	2 (1, 9)	2 (1, 10)	0.441	2 (1, 9)	5 (2, 20)	0.897
TC(mmol/L) Median (IQR)	2.9 (1.2, 5.4)	2.5 (1.1, 4.6)	0.471	3.4 (1.4, 5.8)	3.4 (1.4, 5.3)	0.555
GLU(mmol/L) Median (IQR)	7.00 (6.50, 7.80)	7.20 (6.70, 7.93)	0.314	7.00 (6.50, 7.85)	7.18 (6.70, 7.93)	0.874
TG(mmol/L) Median (IQR)	1.29 (1.03, 1.73)	1.32 (1.06, 2.33)	0.001	1.24 (1.01, 1.60)	1.32 (0.98, 2.70)	0.005
TyG Median (IQR)	8.87 (8.68, 9.15)	8.90 (8.70, 9.42)	0.008	8.85 (8.67, 9.10)	8.87 (8.66, 9.47)	0.035

Data are presented as means ± standard deviations, medians with interquartile ranges, or frequencies and percentages

Abbreviations: *BMI* Body Mass Index, *WWI* Weight Adjusted Waist Index, *HT* Hypertension, *SBP* Systolic Blood Pressure, *DBP* Diastolic Blood Pressure, *PP* Pulse Pressure, *ALT* Alanine Aminotransferase, *AST* Aspartate Aminotransferase, *CR* Creatinine, *NEU* Neutrophils, *LYM* Lymphocytes, *PLT* Platelets, *HDL-C* High-Density Lipoprotein Cholesterol, *LDL* Low-Density Lipoprotein, *TC* Total Cholesterol, *GLU* Fasting Glucose, *TG* Triglycerides, *TyG* Triglyceride-Glucose Index

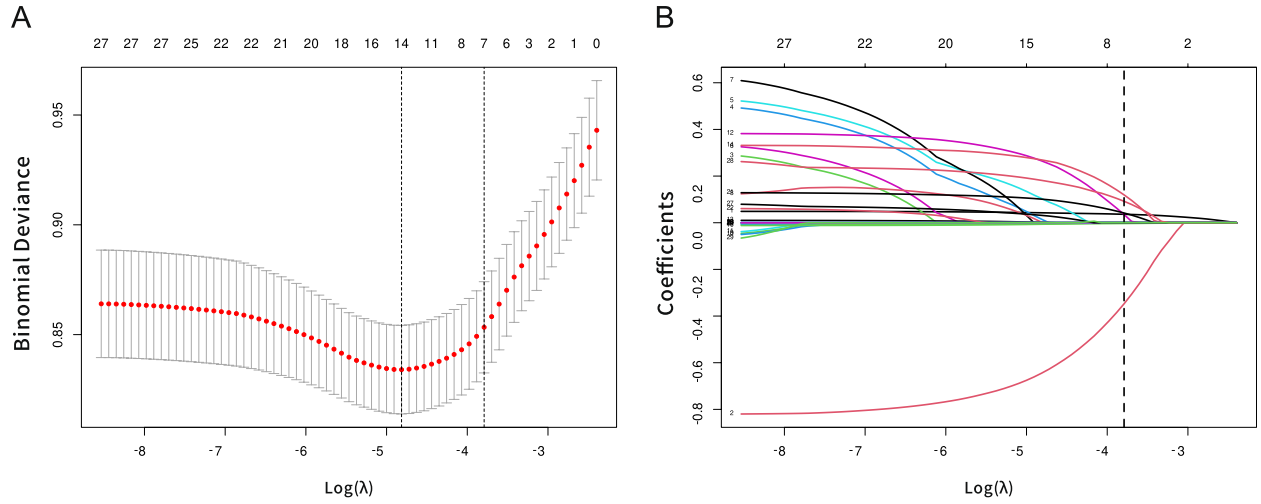


Fig. 2 The process of selecting key potential predictors using Lasso regression analysis. **A** Lasso cross-validation plot for regression. **B** Lasso regression variable selection path (λ = 0.0225701180533294). The least absolute shrinkage and selection operator (Lasso)

Table 3 Results of univariate logistic regression

Characteristic	N	Event N	OR	95%CI	*P*-value
Age	1,499	270	1.06	1.05, 1.08	< 0.001
Gender					
Male	808	185	—	—	
Female	691	85	0.47	0.36, 0.63	< 0.001
HT					
NO	421	51	—	—	
YES	1078	219	1.85	1.33, 2.57	< 0.001
WWI	1499	270	1.50	1.25, 1.81	< 0.001
NEU(1000 cell/uL)	1499	270	1.09	1.02, 1.17	0.009
PLT(1000 cells/uL)	1499	270	0.99	0.99, 0.99	< 0.001
TG(mmol/L)	1499	270	1.18	1.08, 1.30	< 0.001

Abbreviations: *OR* Odds Ratio, *CI* Confidence Interval, *HT* hypertension, *WWI* Weight Adjusted Waist Index, *NEU* Neutrophils, *PLT* Platelets, *TG* Triglycerides

Table 4 Results of multivariate logistic regression for the training cohort

Characteristic	N	Event N	OR	95%CI	*P*-value
Age	**1,499**	**270**	**1.05**	**1.03, 1.07**	**< 0.001**
Gender					
Male	**808**	**185**	**—**	**—**	
Female	**691**	**85**	**0.43**	**0.31, 0.60**	**< 0.001**
HT					
NO	**421**	**51**	**—**	**—**	
YES	**1078**	**219**	**1.50**	**1.06, 2.13**	**0.024**
WWI	**1499**	**270**	**1.47**	**1.17, 1.84**	**< 0.001**
NEU(1000 cell/uL)	**1499**	**270**	**1.15**	**1.06, 1.25**	**< 0.001**
PLT(1000 cells/uL)	**1499**	**270**	**0.99**	**0.99, 1.00**	**< 0.001**
TG(mmol/L)	**1499**	**270**	**1.27**	**1. 14, 1.41**	**< 0.001**

Abbreviations: *OR* Odds Ratio, *CI* Confidence Interval, *HT* hypertension, *WWI* Weight Adjusted Waist Index, *NEU* Neutrophils, *PLT* Platelets, *TG* Triglycerides

both the training set ($X^2 = 7.8519$, df = 8, $p = 0.4481$) and the validation set ($X^2 = 13.056$, df = 8, $p = 0.1099$). This suggests that the differences between the predicted and observed values are not statistically significant, and the model fits well.

Discussion

This study analyzed data from 2,141 DM patients and identified 7 factors as significant risk factors for DM-related CHD (Age, Gender, HT, WWI, NEU, PLT, and TG), based on which a clinical risk prediction model was developed. The AUC of the model was 0.758 in the training set and 0.747 in the validation set, indicating good discriminatory ability and clinical predictive performance. This model fills a gap in research regarding risk prediction tools in the cardiovascular field. Our research is the first to systematically evaluate WWI alongside traditional risk factors (Age, Gender, HT, NEU, PLT, TG) in predicting CHD among diabetics. This innovative approach provides new insights into CHD comorbidity detection in diabetic populations.

This study confirms that age is a significant predictive factor for the development of CHD in DM patients. The risk of CHD in diabetic patients increases markedly with advancing age, consistent with epidemiological evidence showing an age-dependent prevalence of CHD among diabetic individuals [11]. Age-related increases in cardiovascular risk involve multiple mechanisms, including oxidative stress imbalance, endothelial dysfunction, and vascular aging. Older individuals are more susceptible to oxidative stress imbalance, characterized by increased oxidative mediators and decreased antioxidant factors [8]. With aging, the expression of endothelial nitric oxide synthase (eNOS) is downregulated, accelerating the degradation of nitric oxide (NO) [12]. This process causes

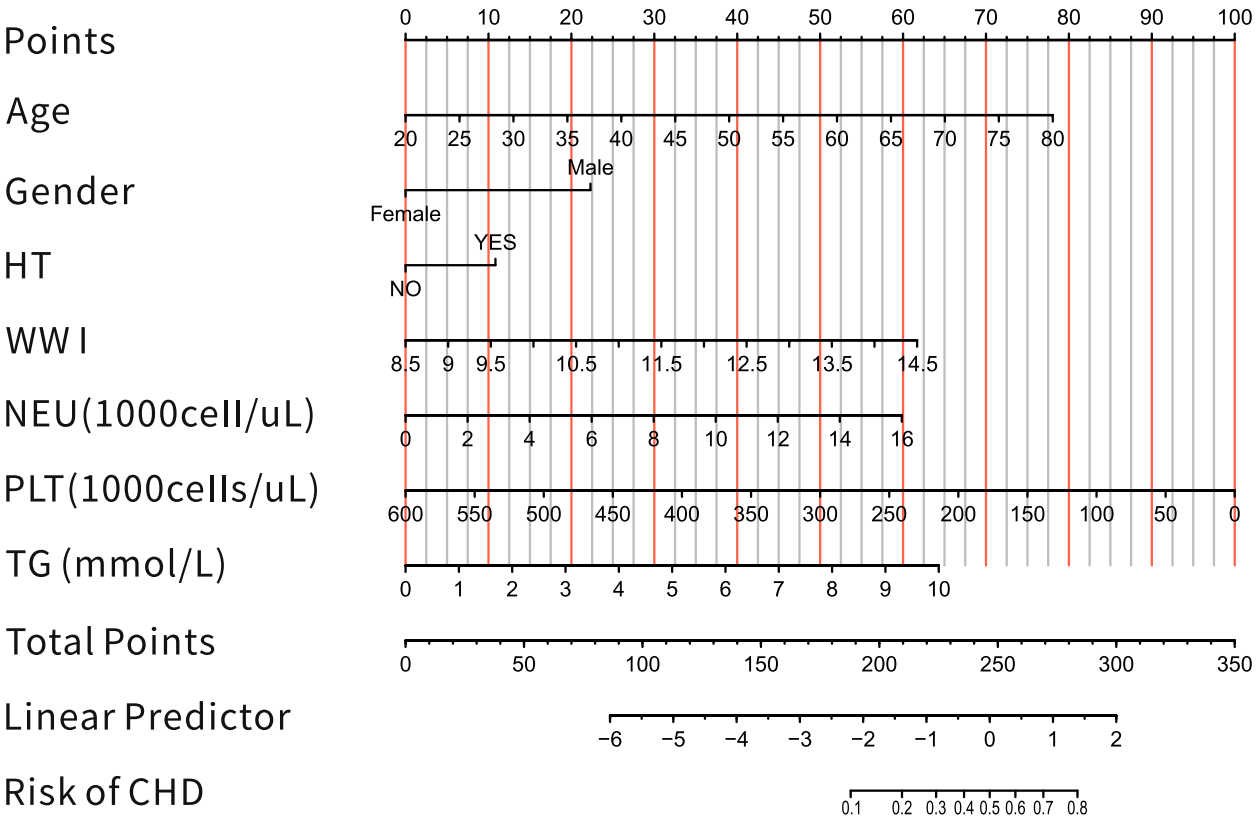


Fig. 3 Nomogram predicting CHD in DM

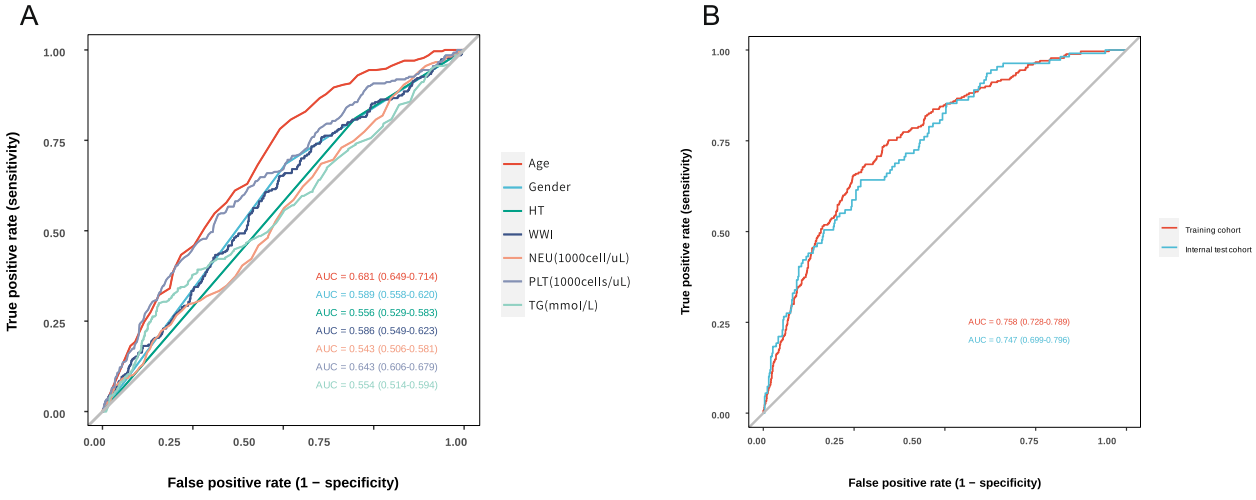


Fig. 4 **A** Training set ROC curve and internal validation set ROC curve. **B** Evaluation of ROC curves for all factors selected. hypertension (HT), Weight Adjusted Waist Index (WWI), Neutrophils (NEU), Platelets (PLT), Triglycerides (TG)

local endothelial dysfunction, contributing to impaired coronary artery function [13]. Furthermore, the hyperglycemic state combined with physiological vascular aging accelerates atherosclerotic plaque formation in elderly diabetic patients, significantly increasing CHD risk [14, 15].

Consistent with previous findings, female diabetic patients have a lower risk of developing CHD compared to males [16, 17]. Estrogen modulates vascular homeostasis via the PI3K/Akt-eNOS signaling pathway, facilitating NO production, inhibiting reactive oxygen species (ROS) generation, and reducing pro-inflammatory cytokines. This mechanism provides protection to endothelial and smooth muscle cells in the coronary artery wall [18–20], thereby reducing cardiovascular risk. However, some studies indicate that diabetic females may have a higher

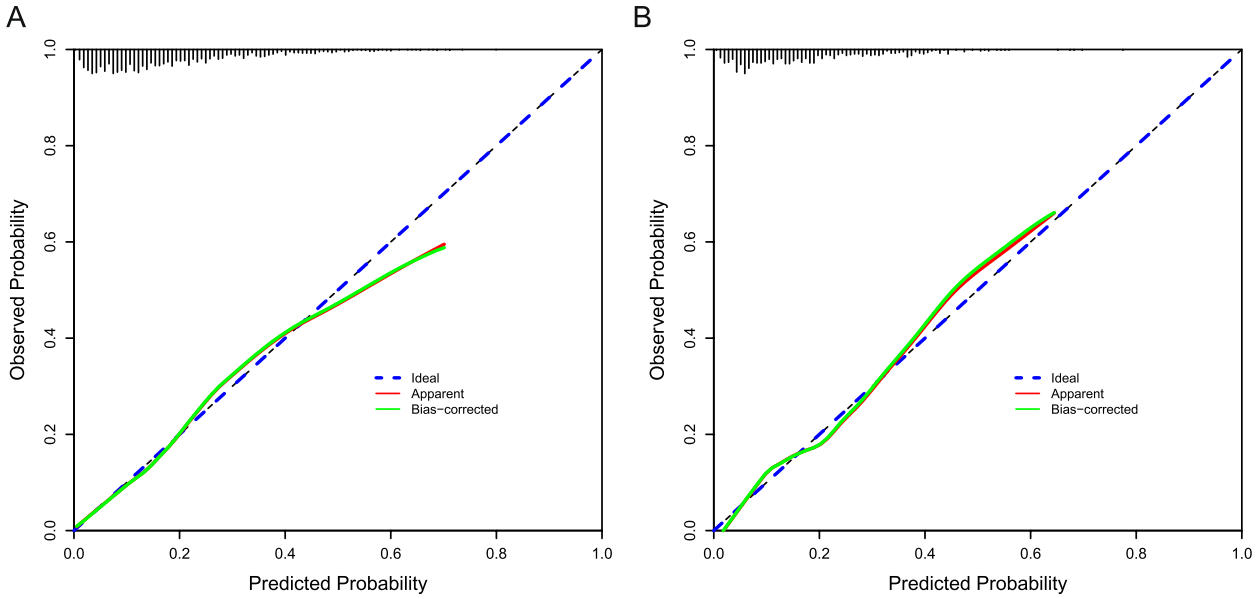


Fig. 5 **A** Training set calibration curves. **B** Internal validation set calibration curves

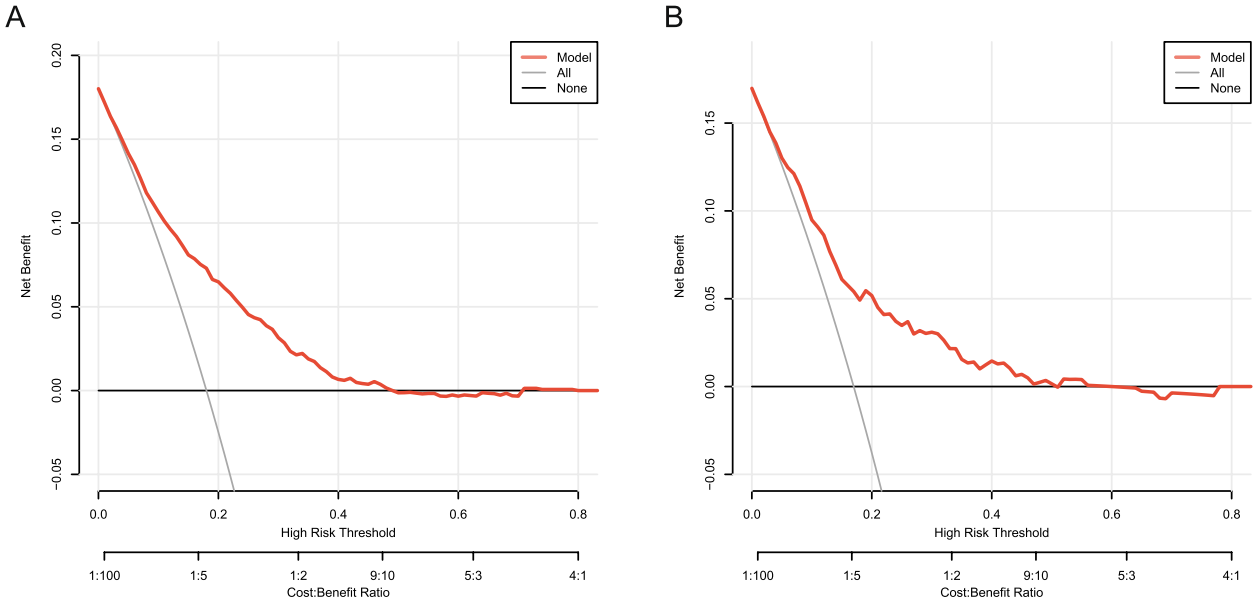


Fig. 6 **A** Training set DCA curve. **B** Internal validation set DCA curve

cardiovascular risk than their male counterparts [21, 22]. This discrepancy could be attributed to demographic differences and diminished estrogen-mediated cardiovascular protection in older women [23]. Additionally, female CHD patients experience non-chest-pain symptoms more frequently than males and are more likely to report associated symptoms such as weakness or pain in the back, shoulder, or neck. Chest pain in women is more frequently diagnosed as Takotsubo cardiomyopathy or coronary vasospasm [18, 24]. This suggests that greater attention should be given to CHD screening and management in postmenopausal female DM patients in clinical practice.

Hypertension is another important predictor of CHD in DM patients. Previous research indicates that DM and hypertension exhibit synergistic additive effects on CHD risk [25]. As an inflammatory condition, hypertension contributes to atherosclerosis and cardiovascular disease [26]. Hypertension directly damages the vascular endothelium through hemodynamic forces and induces endothelial dysfunction by regulating eNOS, thus promoting vascular smooth muscle cell proliferation and luminal narrowing [27–29]. Furthermore, immune cells play a

crucial role in hypertension-driven CHD progression. Studies indicate that hypertension can promote abnormal proliferation and migration of immune cells to target organs, where they release effector cytokines contributing to vascular remodeling [30, 31].

WWI, as a novel obesity assessment index, is positively correlated with CHD risk. Studies have confirmed that WWI can significantly predict the risk of cardiovascular disease and all-cause mortality, with 5-year and 10-year CHD mortality AUC values of 0.734 and 0.757, respectively [32, 33]. Molecular biology research indicates that in obesity, the adipokine leptin activates adrenal glomerular cells through a calcium-dependent mechanism, promoting excessive aldosterone secretion and directly contributing to the progression of cardiovascular diseases [34]. Elevated levels of free fatty acids (FFAs) resulting from obesity-induced adipose tissue expansion can enhance excessive production of ROS, subsequently triggering oxidative stress that induces apoptosis and structural damage in coronary endothelial cells [35, 36]. Additionally, cardiovascular risk factors like DM can further promote eNOS uncoupling and endothelial dysfunction [37], thereby exacerbating oxidative stress [38] and jointly facilitating the development and progression of atherosclerosis. Obesity, serving as a bridge between inflammation and CHD, contributes centrally through mechanisms such as endoplasmic reticulum stress, neutrophil infiltration into adipose tissue, activation of Th1 and Th17 cells, and elevated pro-inflammatory cytokines, collectively promoting chronic inflammation and accelerating atherogenesis [36, 39].

This study reveals the central role of neutrophils in inflammatory regulation in DM-related CHD. Consistent with previous studies [40, 41], we found that circulating neutrophil levels are positively correlated with the risk of cardiovascular complications in DM patients, suggesting their potential as biomarkers for clinical risk prediction. Notably, atherosclerotic plaques in DM patients typically show more extensive inflammatory infiltration and thrombosis [42], leading to a higher susceptibility to cardiovascular events. Specifically, activated neutrophils release ROS, common pathogenic factors for DM and atherosclerosis comorbidity [43], which inhibit NO bioavailability and induce endothelial dysfunction [44, 45]. Moreover, neutrophils secrete myeloperoxidase (MPO), facilitating oxidation of LDL particles and promoting foam cell formation. Elevated circulating MPO levels enhance inflammation and oxidative stress, thereby contributing to coronary artery disease [46, 47]. Additionally, DNA from neutrophil extracellular traps (NETs) binds MPO, inducing endothelial damage and mediating cardiovascular disease progression [48, 49]. Collectively, these mechanisms highlight neutrophils as pivotal effectors linking metabolic disturbances in DM to vascular injury in CHD.

In this research, PLT was negatively correlated with CHD risk in DM patients. Mechanistic studies suggest that enhanced platelet activity in diabetic patients promotes aggregation, subsequently elevating microvascular thromboembolic risk [50]. This pathological change reduces PLT via two distinct pathways: first, platelet activation directly decreases PLT [51]; second, PLT inversely correlate with D-dimer levels [52], implying an increased thrombotic risk when PLT decline due to ongoing platelet consumption during thrombogenesis. Notably, in CHD patients, PLT negatively correlates with mean platelet volume (MPV), with higher MPV associated with increased CHD risk and concurrent PLT reduction [53].

This study highlights the key role of abnormal TG metabolism in cardiovascular complications in DM patients. Previous research confirms abnormal plasma TG levels as primary indicators of lipid metabolic disturbances positively correlated with DM incidence [54], together constituting risk factors for cardiovascular diseases [55]. DM frequently leads to dyslipidemia [56], and diabetic patients with dyslipidemia exhibit a three- to four-fold higher cardiovascular risk than diabetic patients without dyslipidemia [57]. Mechanistically, TG hydrolysis generates high concentrations of lipolytic products, such as oxidized free fatty acids, which induce atherosclerosis through pro-inflammatory cytokines [58]. Additionally, insulin resistance in diabetic individuals may elevate intestinal apoB48 lipoprotein secretion, decrease lipoprotein lipase activity, and enhance hepatic secretion of the VLDL1 subtype, collectively elevating TG levels [59].

The nomogram model developed in this study can rapidly calculate the risk of CHD in DM patients through simple questioning and routine laboratory tests, enabling early risk stratification and individualized intervention. This tool is particularly useful in resource-limited healthcare settings as it does not rely on expensive imaging tests or specialized laboratory diagnostics. By identifying high-risk patients, clinicians can promptly implement preventive measures, such as enhanced lifestyle interventions or pharmacological treatments, to mitigate the progression of CHD.

The limitations of this study are as follows: First, although the NHANES database is representative, the lack of third-party databases or external validation may limit the model's generalizability. Future research should validate the model's performance using other independent databases (e.g., European or Asian cohort studies) or local clinical data. Second, the primary limitation of this study is the reliance on self-reported questionnaire data for outcome measures, which may underestimate the actual prevalence and affect model calibration and accuracy. Future studies should incorporate imaging data

to further enhance predictive capability. Third, Regarding the classification of diabetes, NHANES did not distinguish between type 1 and type 2 diabetes in its open data set, which limited our analysis.In future research, we will actively seek for data sources containing more detailed diabetes classification information, or combine with data from other relevant studies, to make up for this deficiency. Fourth, recall bias may exist due to inaccurate patient reporting, potentially affecting the reliability and accuracy of the findings. Fifth, This study did not include asymptomatic CHD patients, and we consider combining other objective data sources for supplementary verification in future studies to improve the accuracy of disease identification.

Conclusion

This study developed and validated a clinical prediction model for DM patients with CHD, based on multidimensional clinical indicators, including Age, Gender, HT, WWI, NEU, PLT, and TG, using machine learning algorithms. The model demonstrated good calibration performance and clinical net benefit in the validation set, indicating its clinical applicability. These findings contribute to the immediate risk stratification of DM patients with CHD, enabling precise risk stratification and individualized intervention.

Abbreviations

BMI	Body Mass Index
WWI	Weight Adjusted Waist Index
HT	Hypertension
SBP	Systolic Blood Pressure
DBP	Diastolic Blood Pressure
PP	Pulse Pressure
ALT	Alanine Aminotransferase
AST	Aspartate Aminotransferase
CR	Creatinine
NEU	Neutrophils
LYM	Lymphocytes
PLT	Platelets
HDL-C	High-Density Lipoprotein Cholesterol
LDL	Low-Density Lipoprotein
TC	Total Cholesterol
GLU	Fasting Glucose
TG	Triglycerides
TyG	Triglyceride-Glucose Index

Authors' contributions

MW designed the study and wrote the manuscript. MW, WZ and JL collected data. YL and XD analyzed data. YH approved the manuscript. All authors have read and approved the final manuscript.

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Data availability

This study analyzed publicly available data sets. This data can be found at https://www.cdc.gov/nchs/nhanes.

Declarations

Ethics approval and consent to participate

This study adhered to the principles outlined in the Declaration of Helsinki. The NHANES data used in this research were approved by the Institutional Review Board (IRB) of the NCHS. All participants gave written informed consent to participate in this study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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